Q17: Transformer Variations and Alternatives

Map a bottleneck to a design choice

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The question

Q17
How do the latest models vary the transformer architecture, and are there non-transformer approaches?

A family, not one recipe

- Attention, feed-forward paths, residuals, normalization
- Heads/cache, position, activation, routing, and layer layout vary
- Start with the constrained quantity: memory, latency, context, capacity, or quality

Autoregressive decoding

- Every new token consults prior key/value state.
- Multi-query attention shares K/V heads.
- Grouped-query attention is a compromise.

This targets cache traffic; it does not abolish attention.

Position and context

- RoPE encodes relative position in query/key rotations.
- Context length is a configuration and an evaluation question.
- Test distant evidence, distractors, and paraphrases.

Do not conflate sparsities

- MoE: sparse *expert* activation
- Sparse attention: limited token-to-token links
- Quantization: lower precision
- Batching: serving execution choice

State-space alternatives

- Recurrence updates a state while scanning.
- Selective state spaces offer a different linear-time mechanism.
- Test long-range selective recall, not only average benchmark accuracy.

Gu and Dao (2023)

Comparison protocol

1. Match data, token/compute budget, precision, and hardware.
2. Measure quality, latency, memory, and long-context failures.
3. Report which resource, not just which label, improved.

Takeaway

Architecture choice is a hypothesis
State the workload bottleneck and test a design against matched alternatives.